

Lecture 28

4/1/26
JRA

Measuring speed & direction of blood flow using Doppler US

Signal is backscattering (instead of reflection) by RBCs

Finish US, start x-rays

Start w/ premier technique for examining blood flow

US frequency received by doctor is shifted ^{transducer} relative to the frequency that was sent out.

Punch Line:

$$\Delta V = V_{\text{echo}} - V_0$$

ΔV = frequency change / shift

$$\Delta V = 2 V_0 \cos \theta \frac{V_b}{V_w}$$

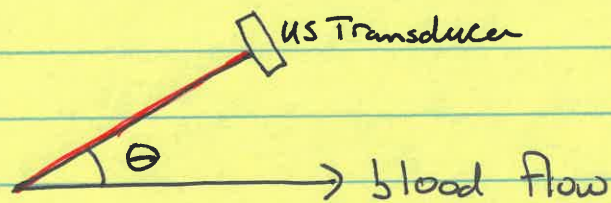
V_0 = original freq (sent out)
 V_b = blood speed
 V_w = wave speed 1540 m/sec

Want to know speed ^{& direction of blood flow} relative to normal
(e.g., is artery blocked)

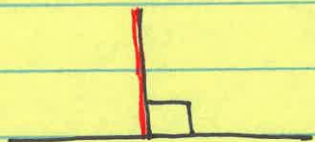
ΔV can be plus or minus

+ moving toward transducer

- " away from "



One bad scenario



$$\theta = 90^\circ$$

$$\cos 90^\circ = 0$$

$\Rightarrow$ No Doppler shift

Simultaneously image so have info pertinent to angle

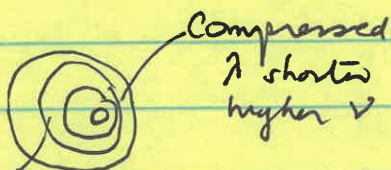
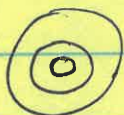
What is Doppler effect - animation

If a source of sound or a receiver of sound is in motion the observed frequency changes

ANIMATION

lon-capr.org doppler

freq w/ which crest
past is frequency



Expanded
& longer
higher v

ANIMATION

Einstein Online

Look at detection rates

Toward	Faster
Away	Slower

Receiver

Toward	Faster
Away	Slower
	Downshifted

In blood flow measurements there are two Doppler shifts

(1) Stationary source (transducer) sends out US
to a moving receiver (RBC)
Hits & backscatters

US is backscattered by Moving source (RBC)
(2) Moving source (RBC)
to a stationary receiver (transducer)

Go thru math in textbook

Moving
receiver

Data Presentation

Superimpose Doppler US data in color coded Doppler data on a B-mode image

BART

Blue away
Red toward

color convention

Brightness encodes magnitude of velocity

dim - slow

bright - fast

Applications {
see PP

Aliasing - false or alias signal from ^{sampling} ~~translating~~
too slowly

Nyquist Shannon sampling theory

Can give wrong direction

VIDEO

Carotid

Daniel J Pyle

Video shows carotid artery to look for blockage

See split in artery

Superimposed B mode & color

Strengths

Noninvasive

Quick, inexpensive

Not harmful

Good for soft tissue

Weaknesses

Strong reflection at A

Attenuation - lose signal

Three more techniques: X-Rays Radiography

Back to EM radiation based imaging

Big Picture
X-rays

Doctors use photons w/ energies suitable for studying body

What is diagnostic X-ray energy regime

15 keV - 150 keV

keV = kilo electron volt = energy unit
suited for
X-rays

$$1 \text{ keV} = 1.6 \cdot 10^{-16} \text{ J}$$

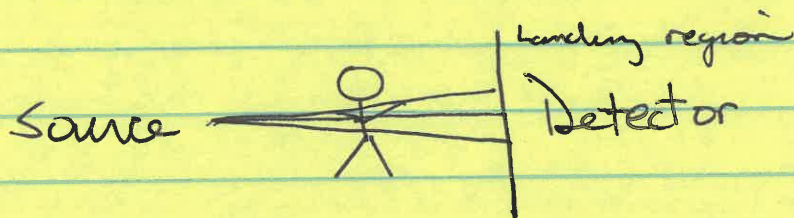
Very high energy units

Chemical bonds are a few eV
So X-ray can break chemical bonds
& ionize atoms

Essence of Radiography

Patient is positioned between an ^{X-ray} source
& a detector

Dental X-ray - put detector in mouth
Source & detector distinct, unlike US



X-rays ideally follow straight line trajectories,
so map onto detector

Bone in chest will absorb ^{some} X-rays & remove them from beam.

Bone & soft tissue

↑ Absorbs more

↑ Absorbs less

Bone absorbs more than soft tissue (e.g. 10x more)

Fewer X-rays behind bone because more removed

More X-rays behind soft tissue because fewer absorbed.

This gives contrast

This creates a shadow of body structures.

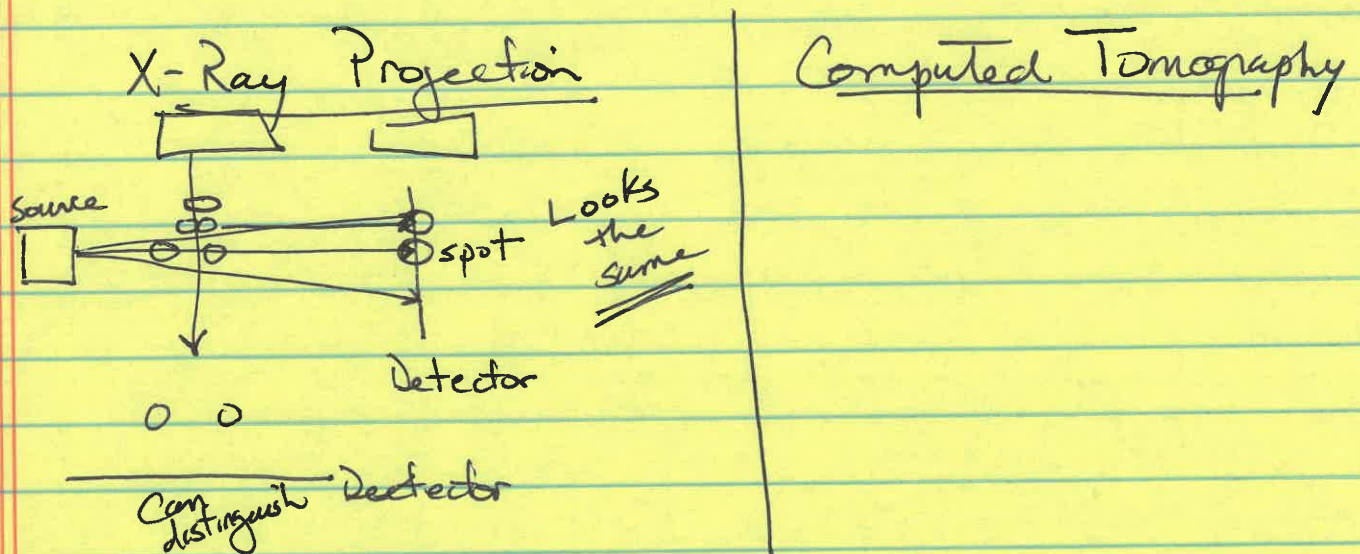
Contrast Flipped in images

← Bone bright
ST dim

Shadow-based imaging

Projection-based imaging

There are limitations on projections



X-ray projections - same image if objects superimposed

Losing depth information - don't know where or how separated

Turn source 90° in example get distinguishable ones

CT take ~ 1000 projections that give complementary info

Turns out you can reconstruct true 3D info correctly

2D patient anatomy $\rightarrow$ 3D

SuperFast trauma assessment

© 2026 Beth A. Scalettar & James R. Abney